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LOAD APPLICATION AND DISPLACEMENT MEASUREMENT TESTING RIG FOR COMMERCIAL AIR BEARINGS

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ABSTRACT

Lithography machines are used in the semiconductor industry to imprint a circuit pattern onto the silicon wafer; the reticle stage holds the reticle that contains this pattern. ASML uses air bearings in their reticle stages to achieve near frictionless motion and micron level tolerances at 20g acceleration. The current air bearings are custom made and costly. The goal of this project is to identify commercial air bearings that meet ASML's specifications and create a test rig that measures air bearing stiffness, i.e. fly height vs. load application, along with flow rate vs. pressure. The team presents load vs. fly height data collected at 60, 80, and 100 psi for the OAV80R air bearing. The collected data was compared to the manufacturer data and the measured fly heights occurred at lower loads than what the manufacturer stated. The team concluded that this is likely due to the large uncertainty created by the measurement flag and suspicions of a damaged air bearing. Future work should include a setup with a flatter flag, such as an optical flat, along with tests involving multiple air bearings of the same type.

PROBLEM DEFINITION

ASML's precision motion machines require air bearings that provide high stiffness and dynamic stability within a limited space. ASML currently manufactures its own air bearings; these custom air bearings are expensive to manufacture due to stringent requirements. The focus of this project is to identify commercially available air bearings that satisfy ASML's specifications and to develop a test rig to quantify their load capacity, fly-height, stiffness, and flow rate.

PROJECT REQUIREMENT AND SPECIFICATIONS

The proposed testing rig is designed to verify the performance of the pre-selected air bearings against the following requirements and specifications:

1. The testing rig must include sensors to measure load, air pressure/flow rate, and fly-height.
2. The displacement sensors shall measure fly-height with less than or equal to 1-micron resolution with a range of at least 0 to 20 µm.
3. The surface supporting the air bearing must have a flatness of less than 2µm over 81 x 150 mm area.
4. The testing set up must be able to replicate the OAV performance graphs and be able to reach 100 psi.
5. The loading application must be able to reach 1650 N to be able to test the bearings at their maximum capacity.

The requirements and specifications for the commercial air bearings are:

Requirements
1. The air bearing test setup shall be able to measure air bearing fly-height at different load and air flow/pressure rates.
Specifications
1. The Moving Frame air bearings shall have a fly-height of 11 ± 2 microns at 1100 N and a minimum fly-height of 6 microns. *
2. The Balance Mass air bearings shall have a fly-height of 14 ± 4 microns at 1650 N and a fly-height range of 6-20 microns.*
3. The Moving Frame air bearings shall have a load capacity of at least 1600 N. **
4. The Balance Mass air bearings shall have a load-capacity of at least 1650 N. **
5. The moving frame air bearing shall have a maximum stiffness of 10E7 N/m with the stiffness at maximum fly-height being 5E7 N/m.***
6. The moving frame air bearing shall fit in a box of 81x150x52 mm [W x L x T]. ****

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7. The balance mass air bearing shall fit in a box of 57x128x22 mm [W x L x T]. ****
8. The test setup equipment and the air bearings bought shall overall not exceed $5000. *****
9. The Moving Frame air bearing cost shall not exceed $4,150 per bearing for a total purchase of four air bearings. *****
10. The balance mass air bearing cost shall not exceed $600 per bearing for a total purchase of ten air bearings. *****
Test Method
* Measurement of displacement via Keyence LK0057 displacement sensors and force via a compressive load sensor. [1]
** As measured using a compressive load sensor.
*** As calculated from Force vs. Displacement curves
**** As measured with micrometers or calipers.
*****As measured by spreadsheet.
Deliverables
1. Testing setup capable of having moving frame and balance mass air bearings tested at different specifications as listed in the specifications document.
2. The purchased air bearings
3. Drawing package of testing setup
4. Technical Report including test data
5. Operations Manual

CONCEPTS

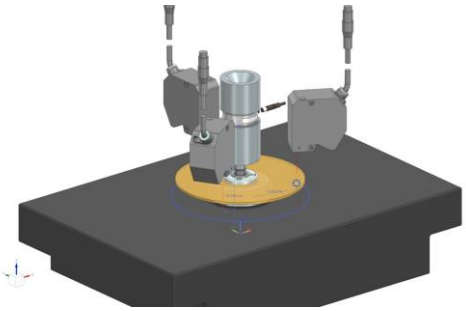


Figure 1 - Gear loaded application

One of the early design concepts that was developed by the team was the gear loaded application method which uses a servo connected to a gear train to apply torque to a rack and pinion. A coupler composed of an inverted cone aligns the rack to apply a force concentric to the air bearing diameter. Between the coupler and air bearing, a compressive load cell measures the force applied. As can be seen in Figure 1, the system uses a cone in cone alignment mechanism to ensure direct load path into the air bearing. Although this design uses a novel alignment mechanism, the overall cost for the load application raised some concerns for the team, and it shifted the focus towards a simpler load application method.

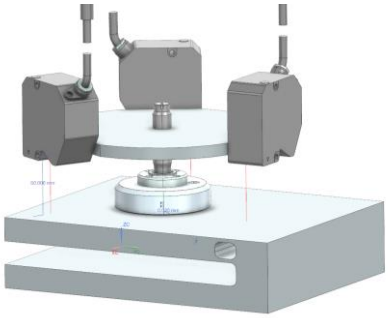


Figure 2 - Wedge and flexure displacement

This testing rig setup would apply a 20-micron displacement and measure the effect it has on force produced by the air bearing. This displacement would be applied by driving a 0.05° wedge into a parallel flexure component. This flexure would ensure that the rising upper plate remains parallel to the lower plate. The wedge would be displaced by a 1/4-100 screw, which compounds the mechanical advantage and allows for the micron level displacement. The air bearing would be fixed in place to a rig that connects back to the bottom testing plate. This ensures that when the wedge moves the top plate upward, the air bearing remains fixed, thus increasing the force. The Keyence sensors attach to the fixed upper portion of the rig and measure the distance to the upper plate of the flexure mechanism. An FEA was conducted on the flexure mechanism to determine if friction forces would cause the displacement to vary across the surface of the loading plate. It was found that the flexure experiences a uniform deflection of 20 microns as expected, and the friction forces contributed negligible amounts of deformation.

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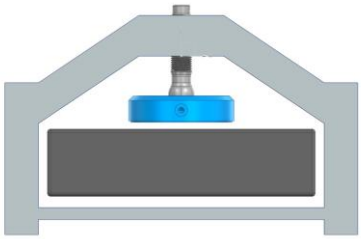


Figure 3 - Rope and pulley load application

As opposed to the gear loaded concept, this concept applies to a downward force below the center of gravity of bearing. If the bearing is left unconstrained, this creates a "self-centering" force that ensures the force application is entirely vertical. This concept requires a yoke that wraps around the granite block, and a frame that elevates the granite surface to make space for the load application method below. Similarly to the gear loaded concept, this idea uses three displacement sensors to define the plane of the air bearing. The initial load application method the team considered was a winch; however, during the preliminary design review, engineers at ASML raised some safety concerns regarding the pully and winch system and pushed the team to brainstorm a way to replace the pully with another mechanism that still uses the top loading bars. Based on their feedback, the team pivoted to a loading bolt, which would tighten to apply load. This concept was ultimately selected due to the advantages of self-centering load application and relative manufacturing ease.

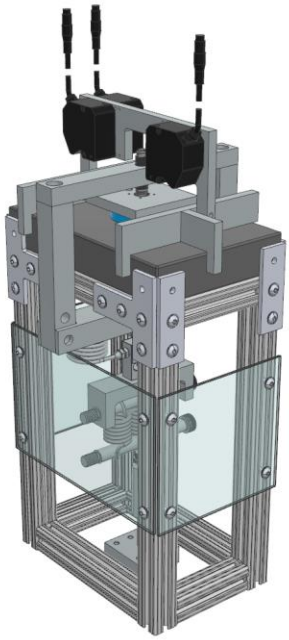


Figure 4 - Final design selection

The figure above shows the final design selected for this project. The system has 5 main sub-assemblies: the loading mechanism, the displacement mount, the safety feature, the granite support table, and the air supply. The loading application uses a self-aligning mechanism through 4 bars attached using shoulder screws. The top bar attaches to the mounting screw which in turn attaches to both the displacement flag and the air bearing. The load is applied by tightening the hex part at the bottom of the table which is attached to the spring holders. As this part tightens, it applies a tension force on the springs which in turn pull on the loading bars. The displacement mount was designed to have a single point of contact on the granite surface. The Keyence sensors are attached to the bridge that connects the two side supports of the mount. Safety panels were also integrated to act as a shield should any component in the bottom loading parts fail primarily in the springs.

MECHANICAL ANALYSIS

Tolerance Analysis:

Location clearance fits and interference fits were used to ensure precise alignment between displacement mounting components, and exact constraint on the granite surface through press fit ball bearings.



Figure 5 - Hole-Slot Alignment Configuration

As shown in the Figure 5 - Hole-Slot Alignment Configuration, two steel dowl pins locate the displacement sensor mount parts precisely with a hole-slot configuration. A ¼-20 bolt provides clamping force between the components. An interference fit (H7/p6) was used for holding the dowel pins and ball bearing. Similarly, the slot was designed as a locational clearance fit (H7/h6) for a snug fit but can be freely assembled and disassembled. These calculations can be found in the Appendix and in the submitted MATLAB code.

Fatigue Analysis:

Fatigue analysis was performed on the steel bar that is directly attached to the bearing, as it experiences significant bending stress. When subjected to multiple cycles, this part is the most likely to fail due to fatigue. The bar is modeled as an 11.9" beam with a rectangular cross-section of 1.25" x 0.875", which is fixed in the middle with a minimum of 0 N and a maximum of

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825 N load acting downwards at each end (based on the maximum load requirement of 1650 N). The maximum bending stress is given by Equation 1.

$$\sigma = \frac{PLc}{2I} \quad (1)$$

P is the 825 N load applied to each end, *L* is the length of the beam (11.9 in), *c* is the distance from the neutral axis of the beam to the top (0.4375 in), and *I* is the moment of inertia of the beam (6.978 x 10^-2 in^4). Using this equation, the maximum stress at the maximum load is 6.91 ksi, and the mean cycle stress is 3.46 ksi. For 1018 steel, the ultimate tensile strength is 63.8 ksi and the yield stress is 53.7 ksi. All factors besides the size factor were assumed to be 1, which was calculated to be 0.895 based on the beam cross-section. The endurance limit is given by Equation 2.

$$S_e' = 0.5 S_{UT} \quad (2)$$

This equation is based on the criteria that the ultimate tensile strength is less than 200 ksi. The endurance limit given the geometric conditions is given by Equation 3.

$$S_e = k_b S_e' \quad (3)$$

k_b is the size factor of 0.895, which yields an endurance limit of 28.55 ksi. Both the Soderberg and Modified Goodman criterion were used to determine if the bar will fail due to fatigue. The conditions for fatigue are given by Equations 4 and 5, which represent the Soderberg and Modified Goodman criterion, respectively.

$$S_e \leq \frac{S_a}{1 - \frac{S_m}{S_y}} \quad (4)$$

$$S_e \leq \frac{S_a}{1 - \frac{S_m}{S_{UT}}} \quad (5)$$

Based on these criteria, the steel bar will not fail due to fatigue.

Fastener Torque Calculation:

Since our main method of load application is through a stud, fastener force and torque calculations were done. To compute the torque needed for an axial force of 1650 N, i.e. maximum load needed according to specifications we use

$$T = K \cdot F_{app} \cdot d$$

K refers to the torque constant and F_{app} refers to the axial bolt force. The torque constant is computed via the following

$$K = \frac{\left(\frac{d_m}{2d}\right)\tan\lambda + f \cdot \sec\lambda}{1 - f\tan\lambda\sec\lambda + 0.625 f_c} = 0.1896 \quad (6)$$

Using this torque constant, the torque needed to reach full loading conditions is 4 N-m or 35.2 in-lbf.

Material Selection:

The team selected 1018 steel for the loading apparatus and flag, and MIC 6 aluminum for the displacement sensor mount. The loading apparatus' four loading bars, two U-shaped spring holders, and the central hex loading component were all manufactured from 1018 steel. While steel is difficult to manufacture, requiring a total of 30 hours of machine time across the mill, lathe, and table saw, its high modulus of elasticity, density, and low cost made it the best material for this application. Steel's high density provided the preload necessary for the self-aligning mechanism while its rigidity allowed for a smaller volume. Furthermore, steel was significantly cheaper than aluminum, which enabled the team to allocate more money to buy other components.

While more expensive, MIC 6 aluminum was selected for the displacement sensor mount due to ease of machining, high flatness and lower density compared with steel. The displacement sensor mount is the most complex part to manufacture, so using MIC 6 enabled for a faster manufacturing timeline. MIC 6 is precision cast aluminum, engineered for ease of machining and comes with its surfaces already flat, meaning it did not need to be faced off, saving significant time. The displacement sensor mount is positioned on top of the granite, high above the table. Out of concern of safety, MIC 6 aluminum's low density meant it would not be as great of a safety hazard to fall.

Spring Sizing Selection:

As explained above, the team used a stud for load applications. Since the load frame is very stiff, it deflects very little, even at maximum load. Due to this, the system reaches the maximum load state in a very small fraction of a turn since the bolt can move down as much as the deflection in the load frame. To have better control over what load is applied, the team decided to add springs between the load frame and the stud to reduce stiffness in the system.

The selection of said springs was determined by their load capacity, stiffness and maximum extended length. The load capacity is the most obvious characteristic since the springs must be able to apply maximum load. Stiffness and maximum length are important in determining our resolution. The ½ - 13 stud used moves down by approximately 0.08 inches per turn. The team decided on a range of 100 - 150 degrees of turning to equal one

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tenth of maximum load. The reason for this was to have enough resolution to easily apply a load setting while also not having to turn too many times. In light of this decision, a spring with a stiffness of 17.51 lbf/mm and a maximum extension of 10mm above the original length was selected.

Once the springs were received, the team tested the stiffnesses in the MTS machine and obtained stiffnesses of 19.4, 19.9 and 20.4 lbf/mm, which were noticeably higher than the rated specifications. The load vs. displacement curves of the springs are shown Figure 6. To accommodate the different rates, a threaded rod end insert was placed between the load cell and the spring holder yoke to allow tilt of the yoke without translating addition load into the load cell.

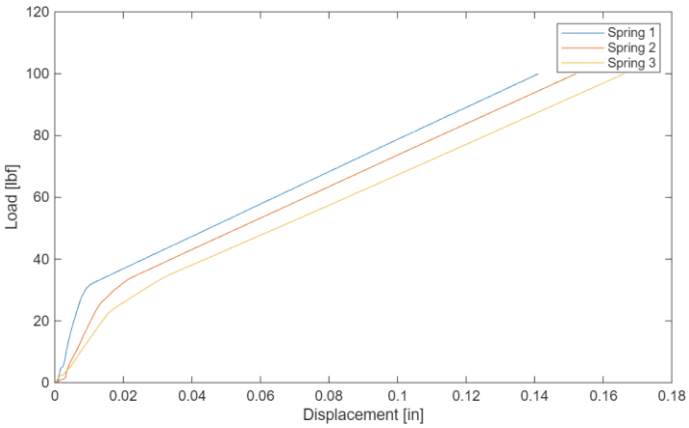


Figure 6 - Load vs. Displacement of the springs as tested by the MTS machine

Finite Element Analysis:

A Finite Element Analysis study was conducted on the loading frame to compute the displacement of the frame at peak loading, and to calculate the stresses within. Figure 2 shows the simulation setup. As can be seen, only half of the structure was modeled, and a symmetric condition was applied at the plane of symmetry. A CTET10 mesh of 0.1-inch element size was generated to represent the 1018 Steel loading frame. The mesh was refined around the bolts to 0.03 inches to better resolve stresses in the region. A bolt preload of 1000 and 1500 lbf were applied to the top and bottom bolts.

Figure 3 shows the results of the simulation. Note that the colormap is on a logarithmic scale. This is to prevent the effect of singularities near the bolted sections overcoming the points of actual high stress.

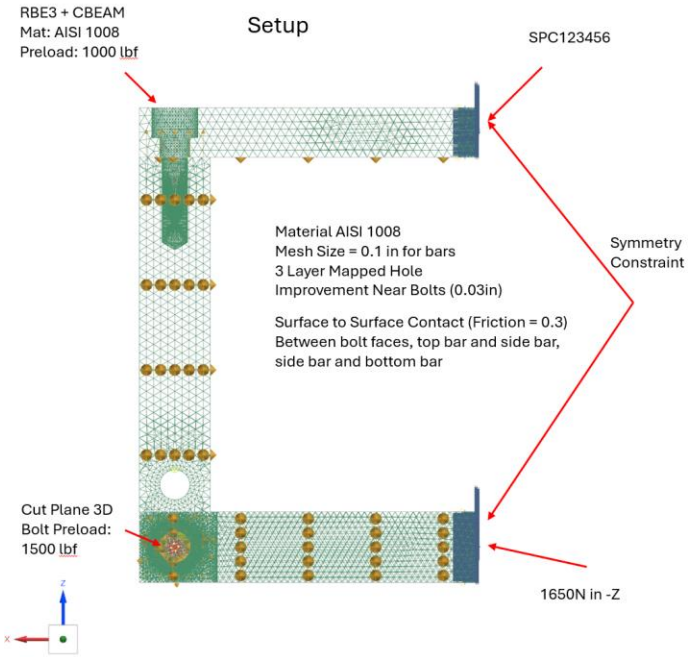


Figure 7 - Setup for Load Frame FEM

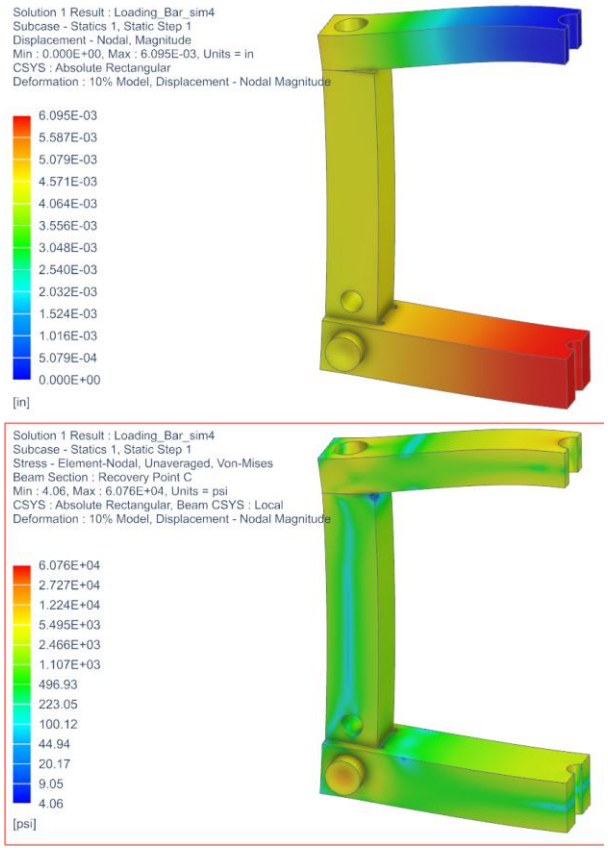


Figure 8 - Load Frame Simulation Results

Fundamental Mechanical Analysis:

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Two fundamental mechanical analysis were made on the testing rig. A stress and buckling calculation on the 8020 column and a flexural stress calculation on the granite surface plate. Both calculations were done in MATLAB.

A stress and buckling calculation were performed on the 8020 columns due to concerns of high stress under the weight of the granite block and applied load. This was done using the following equations:

$$\sigma = \frac{F}{A} \tag{7}$$

$$P_{cr} = \frac{\pi^2 EI}{(KL)^2} \tag{8}$$

Where Equation 1 calculates the normal stress caused by the applied load and Equation 2 determines the buckling load. The calculations determined that the 8020 columns were safe from yield and buckling failure, with a factor of safety of 269.16 and 53.83 respectively.

Due to granite's brittle nature, there was a concern of fracturing during testing. Therefore, the granite's anticipated bending stress was calculated. This was done with the following equation:

$$\sigma_{max} = \frac{Mc}{I} \tag{9}$$

Where *M* is the max moment experienced by the granite, *I* is the moment of inertia, and *c* is the distance from the granite's neutral axis. The anticipated bending stress was calculated to be 240.75 psi. Comparing this to granite's modulus of rupture of 2000 psi yields a factor of safety of 8.3074.

MANUFACTURING

The manufacturing method for this testing rig focused on achieving high structural stability and micron-level precision through a combination of traditional machining and rapid prototyping. For the loading apparatus, the machining process involved cutting to length, drilling precise holes, and tapping threads to ensure a direct load path to the air bearings. Similarly, the displacement flag was upgraded from a thin sheet metal piece to a 0.5-inch-thick steel plate that was ground to 2.5-micron flatness across a 4-inch surface area. The process for manufacturing the flag ensures a reliable datum surface for the sensors, as thin stock often warps during machining.

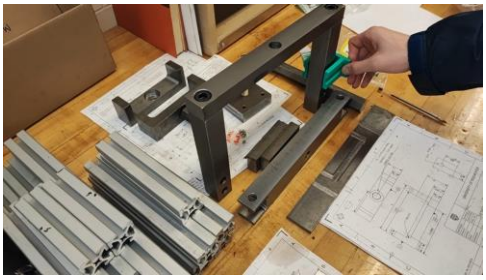


Figure 9 - Loading mechanism components

The displacement mount for the Keyence sensors utilized MIC-6 aluminum for its side supports and bridge due to the material's light weight and ease of manufacturing. The mount required 20 hours of milling to achieve the precise design features and hole/slot locations and tapped features needed for sensor mounting, as well as the installation of the press-fit balls for single-point contact.



Figure 10 - Sensor mount machined components

The flowmeter mounting stand and rotation prevention components were 3D printed using PLA on the Prusa MK4 machine. This method was chosen due to the low cost of 3D printing that doesn't require machine time. These components are under no stress, and for that reason 3Dprinting was the most efficient strategy. The total print time for these pieces was approximately 4 hours.

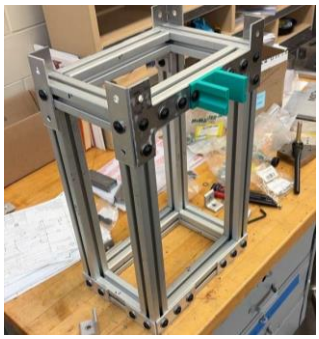


Figure 11 - Assembled 80/20 table

The granite surface support table was constructed using 1.5"x1.5" 80/20 aluminum extrusions. This material was selected for its high strength to-weight ratio and convenient mounting system. The manufacturing process was 8 hours of

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cutting and end-milling to ensure flat mating surfaces for structural stability. The assembled 80/20 is shown in figure 3 above.

Finally, polycarbonate safety panels were machined using both the table-saw available in Rettner shop and the mill. The panels were designed such that they cover the high-risk region with a slight gap to ensure easy access to the hex loading part. The total machine time for these panels was 2 hours. The material choice for these was 1/8 inches thick high impact polycarbonate due to its resistance to impact and transparency.



Figure 12 - Full test rig assembly

A reasonable total cost estimate for this entire project, excluding development time, is approximately $6400 rated at $100/hour. This cost excludes the 3D printing time. Separate from the build cost, the estimated development time for the 5 team members covering CAD modeling, system integration, structural analysis and documentation totals 599.8 hours, resulting in a total development cost of $59,980 as detailed in Table 1.

Table 1 - Purchased Hardware Cost Breakdown

Item Purchased	**Cost**
Ultra Zero Grade Air Tank	$130.00
RMB-52-SSV Flowmeter	$88.25
5/32” OD x 1/4" NPT Male	$10.00
Nylon Tubing 5/32" x 100'	$29.00
4mm OD x M5x0.8 push to connect	$8.01
4mm OD x M3x0.8 push to connect	$5.40
5/32" OD x 1/8" NPT	$12.53
Steel Stock	$36.00
80/20 stock	$345.00
Ball Joint Rod End Inserts	$9.65
Thin-Profile Hex Nuts	$6.54
T-Slotted Framing	$64.20
Ratcheting Combination Wrenches	$23.57
Deep-Offset Box Wrenches	$26.41
Extension Springs	$34.05
1.25" square bar-16"	$25.00
1"x1.25"-12" long	$21.00
1"x3"-3" long	$7.00
1"X3"-3" long	$9.00
1.5"x2"-12" long	$35.00
0.5" 1010 Round Stock-16" long	$4.00
Air bearings	$888.00
Granite surface	$208.00
Load sensor	$398.75
Clear Impact-Resistant Polycarbonate	$44.74
Needle-Roller Thrust Bearing	$7.82
Alloy Steel Shoulder Screw	$13.68
0.063" Thick Washer	$13.12
Black-Oxide Alloy Steel Socket Head Screw	$9.33
Low-Carbon Steel Hex Bar	$13.80
Black-Oxide Alloy Steel Socket Head Screw	$7.97
Alloy Steel Shoulder Screw	$30.40
Total cost	**$2,571.97**

Table 2 - Team member shop time Summary

Team member	**Shop time**	**Cost**
Kaan Aytekin	24	2400
Amel Rouabhia	10	1000
Darren Chen	20	2000
Michael Bartusek	8	800
Ryan Zorzi	2	200
Total	**64**	**$6,400**

Table 3 - Development time per team member excluding shop time starting Jan 22nd

Team member	**Development time**	**Cost**
Kaan Aytekin	132.8	13,280
Amel Rouabhia	120.5	12,050
Darren Chen	115.0	11,500
Michael Bartusek	108.5	10,850
Ryan Zorzi	123.0	12,300
Total	**599.8**	**$59,980**

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If the system were to be scaled to 1,000 units, several changes would be necessary to reduce the high machine time and overall costs. The steel components of the load application subsystem should transition from traditional CNC milling to investment casting, which allows for near-net-shape production and would significantly reduce individual part machining time.

TEST PLAN AND RESULTS

The experimental evaluation of the air bearing performance focused on establishing a precise correlation between applied load and fly height. The goal for testing was to recreate the two published performance diagrams: applied load vs fly height and flow rate vs input pressure. Prior to data collection, both the load cell and the Keyence sensors were calibrated to ensure accurate data collection during the experiment. The processes started with initializing the load cell, which was zeroed while in neutral state. The total applied load during testing was defined as the sum of the yoke's deadweight and the calibrated output from the load cell amplifier. Following the force calibration, a detailed calibration of the three Keyence sensors was performed to ensure perpendicularity to the displacement flag and to obtain information regarding the tilt of the plane formed by the three data points. This was achieved by measuring the positional data between the two gauge blocks of known nominal lengths, which allowed for the calculation of the laser's angular alignment relative to the granite surface. This angular correction was an important step in calibration. The angle between the sensor and the granite surface was computed using Equation 10.

$$cos\theta = \frac{differerence\ in\ no\min al\ length}{difference\ in\ measured\ length} \quad (10)$$

Once the calibration process was complete, data collection was initiated by establishing a reliable baseline before each trial by zeroing the Keyence sensors with the loading mechanism attached and the air supply turned off. This step defined the reference position for zero fly height allowing for consistent measurements and repeatability across all trails. Once this step was complete, the air supply was turned on and set to a pressure of 60 psi for the first set of data, and the initial load and fly height were recorded. The initial load represents the deadweight of the system. Incremental loads were then applied by tightening the hex loading piece, which applied a tension force on the springs in steps of approximately 15 units on the amplifier, corresponding to a physical weight increase of roughly 35 lbs. per increment. The conversion equation from load cell reading to applied load is given by Equation 11.

$$Load_{applied} = (2.5458 \times L_{cell} + 0.1855) + 16.35 \quad (11)$$

As the load was applied at each incremental value, the displacement from each sensor was recorded. A total of 8 data points were collected to ensure a valid comparison with the performance diagrams provided by OAV for the 60 and 80 psi curves. To ensure the highest level of repeatability, a resetting procedure was implanted between every trial. After recording a trial, the hex loading piece was fully loosened to unload the bearing, and the air supply was turned off. After this step was complete, a re-zeroing of the sensors was performed, and data was collected again for the remaining trails. This resetting step was implemented to account for any mechanical settling or flag rotation during the loading cycle. Data was recorded for three pressure values of 60 psi, 80 psi and 100 psi with three trials for each pressure. The flow rate vs pressure input deliverable was measured using an RMB-52-SSV Flowmeter. This flowmeter has a range of 4 – 50 standard cubic feet per hour, which according to the existing performance diagrams should be able to capture results between 40 – 100 psi.

After data collection, the raw data was processed in MATLAB to produce the performance diagrams of flyweight vs displacement at different pressure values. Additionally, the calibration data was plotted to visualize the tilt in the Keyence values.

Collected Data

Fly height data from the three sensors were collected at 8 load increments for 60, 80 and 100 psi. Knowing the XY coordinates of the lasers from our mount and the Z coordinates from the sensor readings, planes for each fly height were generated to represent the flag as shown in Figure 13 - Planes generated from the 3 points read by the sensor. Using the equations of the generated planes, the fly height at the approximate center of the air bearing was computed.

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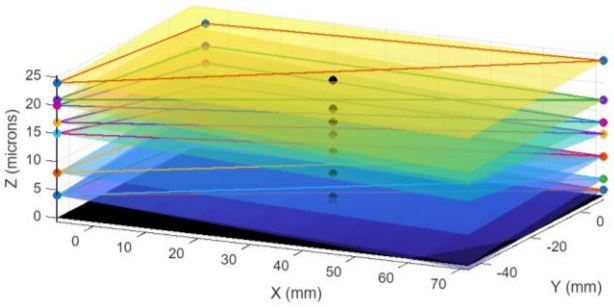


Figure 13 - Planes generated for each load case from the 3 points read by the sensor

The center fly heights were plotted against the applied loads for 60, 80 and 100 psi of air pressure in Figure 14, Figure 15 and Figure 16 respectively. In terms of flyheight measurement, the bottleneck in the uncertainty was the flatness of the flag. The overall flatness of the flag was on the order of 5 microns as measured by a dial gauge on a surface plate. Locally, that is within a ¾ inch radius around where the displacement sensors read, the flatness was around 2 microns. The remainder of the sensor uncertainty was combined with the flag flatness by RSS.

For the 60 and 80 psi case, the manufacturer has published load vs. flyheight data which are plotted along with our results. Unfortunately the collected data does not reflect the same characteristics as the manufacturer specifications.

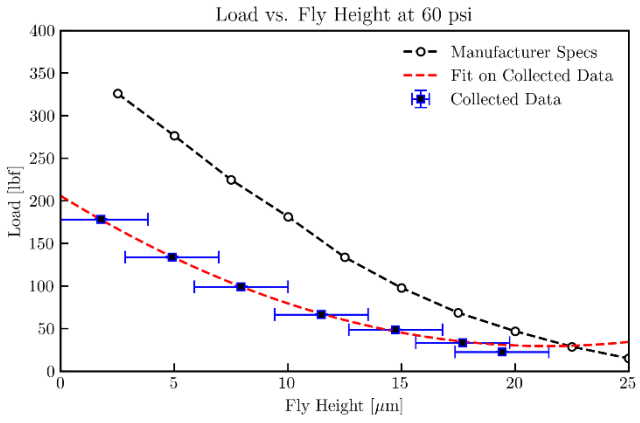


Figure 14 - Load vs, Fly height data for 60 psi air

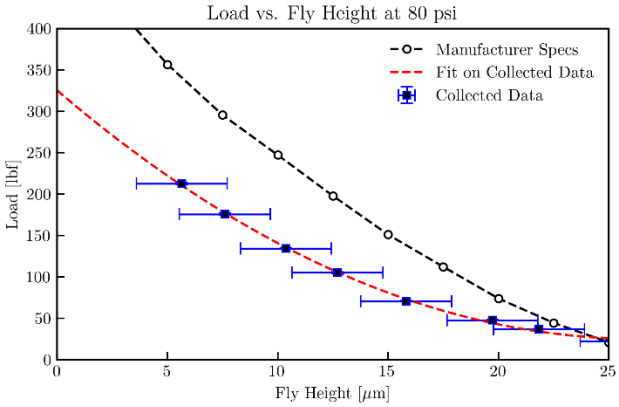


Figure 15 - Load vs. Fly Height data for 80 psi air

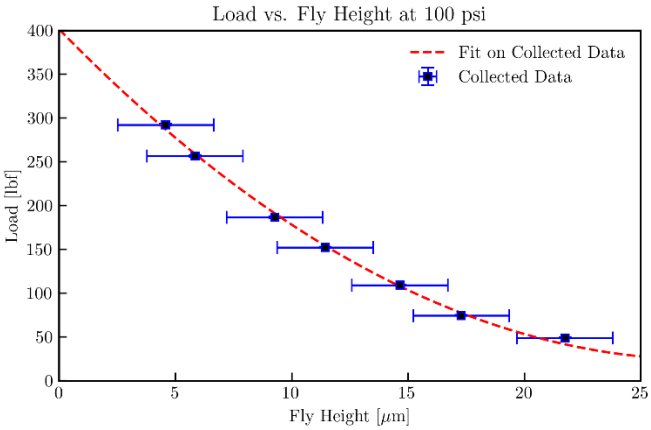


Figure 16 - Load vs. Fly Height data for 100 psi air

The results for the flow rate vs input pressure delivery were concerning, as even when tested at 100psi, the flowmeter never exceeded a reading of 8 SCFH. This is considerably inconsistent with the expected results. This problem could have several causes: a faulty flowmeter, pressure drops in the tubing, pressure drops in the flowmeter, or a faulty air bearing. The tubing between the regulator output and the flowmeter input is thicker than the tubing between the flowmeter output and air bearing input, so it is unlikely that pressure drops before the flowmeter are the problem. The next assumption was that the tubing between the flowmeter output and the air bearing was too narrow and causing pressure drops and stagnation. To test this, the tube was disconnected from the air bearing, allowing the air to exit the tube freely at 30 SCFH. While the air continued to flow, the tubing was removed from the flowmeter output, which immediately led to the flow rate increasing to 40 SCFH. This would seem to suggest that the tubing is reducing the flow rate;

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however, a second test at a more realistic flow rate of 12 SCFH only increased the flow rate to 13 SCFH. Since the tubing restriction is only substantial at higher flow rates, it would fail to explain why the flow fails to exceed 5 SCFH when it is expected to surpass 15 SCFH. A minor pressure drop in the flowmeter would be expected, yet that does not explain the extreme reduction in flow. Once eliminating these variables, the epoxy bonded bearing was substituted in, and produced far higher flow rate readings, even at moderate pressures. This led the team to believe that the flow rate issue is an internal issue with the 80 mm bearing itself, which the diagnostic tools available are generally incapable of assessing or quantifying. Given this setback, the team was unable to complete the delivery of the flow rate vs input pressure performance diagram.

INTELLECTUAL PROPERTY

This design and operating procedure are patentable because this device is a novel approach to measuring air bearing stiffness. There have been similar attempts at developing equipment for measuring air bearing stiffness; however, they offer a different method of load application and fly-height measurement. Notably, the Shanghai Microelectronics Equipment (SMEE) company created an apparatus that loads the air bearings from the top with coarse and fine adjustment screws [2]. Their design, however, fails to address sideways loading because the air bearing is fixed in the X and Y directions. Furthermore, their design is prone to bending due to long screws used to apply the load.

SOCIETAL AND ENVIRONMENTAL IMPLICATIONS

By allowing ASML to switch from expensive custom-made air bearings to cheaper commercially available options, ASML will be able to reduce the price for their machines. This is valuable to them, as the price of their lithography machines can be prohibitively expensive. On April 23rd, semiconductor manufacturer TSMC announced that they would not be purchasing ASML's latest line of machines due to cost concerns [3]. This announcement led to ASML's stock falling by 3%.

Moreso than simply earning ASML more money, allowing them to sell more machines will result in more microchips being manufactured. The recent explosion in data centers to accommodate large-scale AI usage has drastically increased the need for semiconductors [4]. Improving accessibility to these semiconductors would have widespread positive effects.

RECOMMENDATIONS FOR FUTURE WORK

Overall, the team is confident that the current design is the strongest of the considered concepts for force application and displacement measurement. The self-centering approach has proven effective at keeping the load application strictly vertical, and the Keyence sensors are far superior to any other displacement measurement device within the budget. There are still several modifications that could be implemented to improve the testing rig .

The most pressing shortcoming of the experimental results is the flowmeter readings; however, it is not immediately clear what can be done to solve this even with more time available. If the flow through the bearing is in fact as low as the readings suggest, a simple solution would be ordering a second flowmeter that measures flow between 1 and 10 SCFH. This outcome would directly contradict OAV's published performance diagram. The experiments do show a lower load capacity compared to the expected results, which may suggest the bearing is damaged in a way that limits both air flow and output pressure. Given these concerns, purchasing another 80mm round air bearing would be a better next step than altering the flowmeter.

Additionally, incorporating the ability to test epoxy bonded air bearings would be a major improvement to the current system. Since permanently bonding an air bearing for a performance test is impractical, a constraint would have to be manufactured that can serve the same role as the mounting screw on non-epoxy bonded air bearings. Ideally, this mounting device would be adjustable so it could incorporate different dimensions of epoxy bonded bearings.

Regarding modularity, the flag design currently mounts to the 80mm round bearing using 4 holes meant to attach the mounting screw. If a bearing didn't have those holes or had an extrusion on its top surface, the current flag would not be applicable and ASML would have to manufacture a new one to test that bearing. The flag is a challenging part to manufacture due to the extreme flatness requirement, so the design of a customizable flag system that accommodates different bearing geometries would be a major improvement. The testing rig already includes mechanisms that allow testing of different bearing geometries, such as secondary holes on the yoke and multiple slots for the Keyence sensors on the displacement mount. With an epoxy bonded bearing mount and a customizable flag, the testing rig would be fully able to test a wide range of bearings with minimal effort required by the user.

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We would like to thank our sponsor Ognjen Bosic for his guidance on this project, ASML, for accommodating our tour of their facility and sponsoring this project, and the Department of Mechanical Engineering for sponsoring our trip to ASML and funding this project. We would also like to thank Professor Muir for his invaluable technical advice, Christine Pratt for assisting the team with the Keyence sensor operation and air flow consulting, Professor Mohammad for his time assisting with our load cell circuitry, Bill Mildenberger for manufacturing the flag and

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assisting with machining operations, Sam Kriegsman for an incredible amount of machining advice, Jim Alkins for assistance in the Rettner machine shop, Ed Herger for advising us on displacement measuring techniques, Mason Garlatti and the Chemical Engineering Department for lending us the regulator and assisting with the air supply tubing, Arthur Chen with assistance during setup, and our TA Matthew Hook for helping us in our biweekly meetings.

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APPENDIX A

One of the main deliverables was a list of air bearings that function for ASML's specifications. Below is a list of air bearings that the team was able to find. While we identified the three best air bearings, the rest of the list serves as useful information for designs in the future.

Name	Bearing Type	Cost (USD)	Mass (g)	Ideal Load (N)	Stiffness (N/m)	Fly Height (µm)	Input Pressure Limit (psi)	Length (mm)	Width (mm)	Height (mm)	Volume (mm^3)
Ideal MF		4150	575	1600	1.00E+08	6	100	150	81	52	631800
S125001	Porous Media		295	1112	1.10E+08	5	60	100	50	25	125000
OAVBB70150	Epoxy Bonded	718	320	2846	9.70E+07	5	100	150	72	15	162000
OAVF50L100	Porous Material	391	295	1184	9.70E+07	5	100	100	50	25	125000
OAVR080R	Porous Material	414	231	1189	1.17E+08	5	100	80	80	20	100531
S127501	Porous Media		1411	2580	3.24E+08	5	60	150	75	50	562500
S1010001	Porous Media		440	1776	1.75E+08	5	60	100	100	25	196350
OAVBB38118	Epoxy Bonded	474	129	934		5	100	118	38	12.5	56050
A-411.080	Discrete Orfice		259	1010	1.11E+08	12.5	80	80	80	20	100531
G17072115 - HXX	Bonded		164.5	1290				71.7	71.7	15	77113
G17072215 - HXX	Bonded	802.44	337.5	2580	1.11E+08	5	100	146.7	71.7	15	157776
S108001	Porous Material		240	1110	1.14E+08	5	100	80	80	20	100531
LRAP300	Porous Material		224	1320	1.29E+08	5	100	76.2	76.2	19.8	90295
LRAP350	Porous Material		346.7	1791	1.77E+08	5	100	88.9	88.9	22.6	140282
LSAP200400	Porous Material		316.1	1408	1.58E+08	5	100	101.6	50.8	24.4	125935
A-411.100	Discrete Orfice		505	1710	1.75E+08	12.5	80	100	100	25	196350
EZ-0053-080	Discrete Orfice		240	1800	4.00E+08	3.9	87	80	80	26	130690
EZ-0072-50x100	Discrete Orfice		368	1650	2.50E+08	3.8	87	100	50	30	150000
EZ-0075-50x100	Discrete Orfice		335	1650	2.50E+08	3.8	87	100	50	29.5	147500

Figure 17 - Table of selected air bearings that fit in ASML specifications

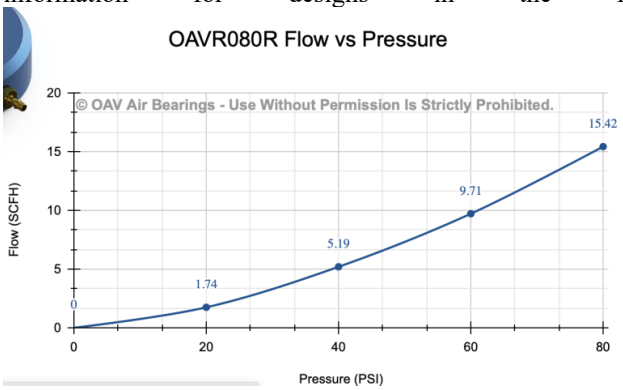


Figure 18 - Flow vs. Pressure graph for the OAV080R air bearing

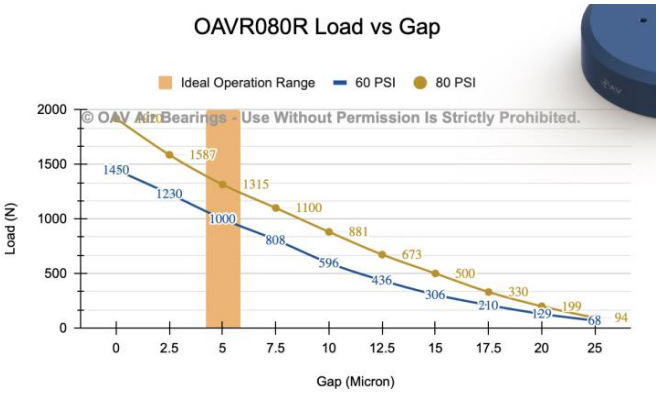


Figure 19 - Load vs. Gap graph for the OAV080R air bearing

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Figure 17 - Table of selected air bearings that fit in ASML specifications

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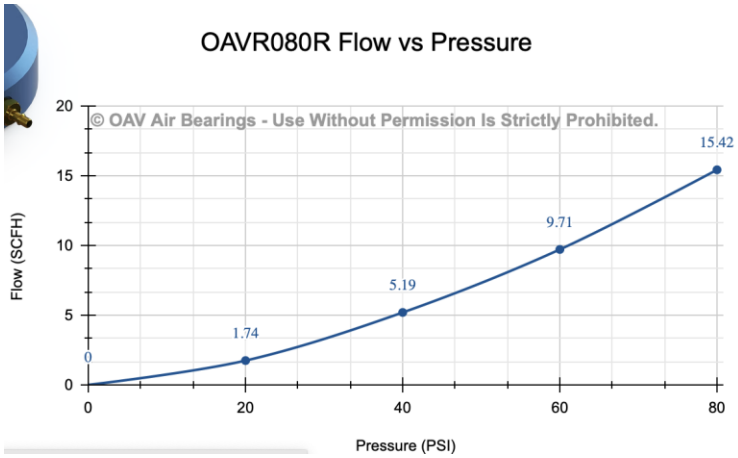


Figure 18 - Flow vs. Pressure graph for the OAV080R air bearing

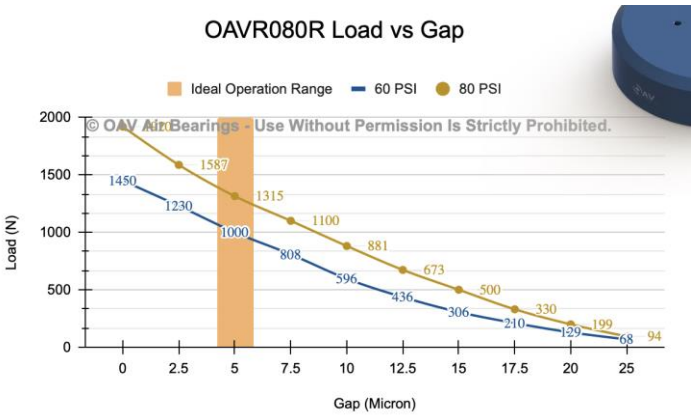


Figure 19 - Load vs. Gap graph for the OAV080R air bearing

APPENDIX B

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MATLAB DATA PROCESSING

```
% Fundamental Mechanical Analysis
%% Stress and Buckling Analysis of 8020 Part No. 1515
clear all; clc

IX = 0.2542; % in^4, moment of inertia X
IY = IX; % moment of inertia Y
A = 1.152; % in^2, cross section area
Y = 25000; % psi, yield strength
E = 10000000; % psi, Young's modulus

Gw = 28; % lbf, granite weight
F = 400; % lbf, max anticipated force applied
totF = Gw+F; % lbf, total force anticipated

S = (Gw+F)/4/A; % psi, compressive stress
FS = Y/S; % f.s of 8020

L = 16.5; % in, length of table legs
K = 1; % pin-pin effective length factor
Pcr = pi^2*E*IX/(K*L)^2; % buckling load
FScr = Pcr/(Gw+F)/4; % f.s of buckling

disp("8020 column max compressive stress is: "+S+"psi")
disp("The yield strength of 6061-T6 is:"+Y+"psi")
disp("The max load on the 8020 column is: "+totF+"lbf")
disp("The buckling load is: "+Pcr+"lbf")
disp("The compressive F.S is: "+FS)
disp("The buckling F.S is: "+FScr)

%% Granite Block Stress Analysis

l = 12; % in, length of granite block
w = 8; % in, width
t = 2; % in, thickness

c = t/2; % in, distance from neutral axis
M = (F+Gw)/2*l/2; % lbf*in, max moment on block
I = w*t^3/12; % in^4, moment of intertia
Smax = M*c/I; % psi, max bending stress experienced by granite, eqn simplified as: 3*(F+Gw)*l/2/w/t^2

Sr = 2000; % psi, modulus of rupture (https://www.matweb.com/search/datasheet.aspx?matguid=a549b92993f640d293329393e59bca30)

FSgranite = Sr/Smax; % granite F.S

disp("The max bending stress experienced by the granite is: "+Smax+"psi")
disp("The modulus of rupture is: "+Sr+"psi")
disp("The F.S of the granite is: "+FSgranite)

%% Tolerance Analysis
basicSize = 0.25; % in
% pin location feature (H7/h6)
deltaD = 0.0006; % in, Shigleys (H7)
Dmin = basicSize % in, locating hole min
Dmax = basicSize + deltaD % in, locating hole max

delta = 0.0003; % in, pin deviation (+) from McMaster
deltad = 0.0004; % in, pin tolerance grade (h6)
dmax = basicSize+delta % in, pin max size
dmin = basicSize-deltad % in, pin min size


% pin pressfit (H7/p6)
deltad2 = 0.0006; % in, pin tolerance grade (p6)
dmax2 = basicSize+delta % in, pin max size
dmin2 = basicSize-deltad2 % in, pin min size
```


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